Vertical Jump Coach Agent — Progress Report

WatSPEED Agentic AI · Course Project · Sean Tang · due Nov 8, 2026
Code: github.com/seantanger/thp-jump-coach-agent (branch `main`)

For this project, I am building an AI coach for vertical jump training based on the information from John Evans / THP Strength YouTube video transcripts. The work I have done is with the help of Claude Code. The agents use the claude API. The first thing I did was collect all of the transcripts from YouTube using yt-dlp, and preprocess the text with regex rules. I then tried three chunking techniques: fixed-size character splitting (LangChain's `RecursiveCharacterTextSplitter`) at two chunk sizes, also a method that cuts the transcript into ~60-second windows at natural pauses in speech. After embedding the chunks using nomic embed in Ollama, I made a simple LangGraph loop where the Coach calls a search tool itself (up to 4 searches, rewriting its query if needed) and answers with the video and timestamp. I compared each of the three chunking methods with three search methods (keyword/BM25, vector, and a hybrid of the two) on 30 test questions. The best combination was the larger character chunks with vector search. I also built a second agent, a Planner, that writes multi-week training programs from reviewed templates, runs them through 8 code checks, and waits for my approval before saving.

I went through 4 versions of the Coach prompt, most recently to make answers more concise. For safeguards, a code check rejects any answer that cites a source the search didn't return, a third agent (the Verifier) checks each point against its source in a fresh context, and a code safety check sends messages with red-flag pain symptoms (like a pop in the knee) to a physio instead of coaching them.

Next Steps:

- Plan quality: reading the first plan, sessions need more detail. Accessories should be named exercises ("general lower-body accessory 3×10" doesn't say which one), every lift needs a load (month-1 power cleans have no % because the transcript doesn't give one), and sessions need warm-up and ramp-up sets instead of one weight for every set. I also need to still work on the re-editing plan process. E.g allow the user to keep reiterating the plan if they have suggestions.
- Connect the Planner and Coach. Right now they are two separate LangGraph graphs: a code router detects plan requests, but the chat doesn't pass them to the Planner yet. I want to combine them into one parent graph with the Coach and Planner as subgraphs. Should routing stay in code or use an LLM?

- Give the Coach access to the current plan and profile, so its answers fit where the athlete is in the program.
- Memory: right now the profile and each plan version are saved as JSON files. Would something like Letta be better?
- Training log: record sessions as done, changed or missed, plus pain and jump results, and use them to decide when the athlete moves to the next stage. (Not sure if this should be like a UI or the user should enter a chat to tell the system that they have finished a workout)
- Optional: MCP server, workout variety in the Planner.

Demos

These call the Claude API (about 4¢ for the Coach question, about 25¢ for a plan) and read the transcript index. The outputs are saved in this notebook. The embedding model used is nomic embedding model in Ollama.

In [1]:
```
import os, sys
from pathlib import Path

ROOT = (Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd())
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
os.environ.setdefault("RETRIEVAL_MODE", "dense")  # dense (vector) search ne
```

Out[1]:
```
'dense'
```

The Coach's LangGraph setup (`src/coach/graph.py`). Two nodes: `coach` (the LLM with tools bound) and `tools` (runs the searches, capped at 4). The graph loops coach → tools → coach until the LLM stops calling tools. `answer_question` (used below) builds this graph, runs it, checks the citations, and sends the answer to the Verifier.

1. Ask the Coach a question. The Coach searches the library, answers with citations, then the Verifier checks it. (The two agents are linked through code, not the LangGraph)

In [2]:
```
from src.coach.qa import answer_question

r = answer_question("Should I static stretch before jumping?")
print(r["compact"])
print("\nsearches:", r["queries"])
print("citation check passed:", r["check"]["passed"], "| Verifier:", r["chec
```

```
Generally no. Skip static stretching before jumping and use dynamic flexibil
ity instead. If you do need static stretches, for example because of very ti
ght ankles or hips, do them after the warm-up and power work.

- Don't do static flexibility before training, because it can decrease force
output a little. It's only worth considering if you're so immobile that you
can't reach squat or lunge positions. [1]
- Long, deep static holds before jumping can fatigue the nervous system, and
the coaches say that if you do long holds and then jump, you're going to be
in trouble. [2]
- Warm up with skips, drills and low-level bounding. Then do dynamic flexibi
lity such as toe touches, deep bodyweight squats, long lunges and side squat
s. Then do progressive jumps starting around 10% intensity. [3]
- If you want static stretching, do it after the warm-up. For example, hold
a 2-minute ankle stretch with a plate on the knee, or a banded hip stretch,
to make squatting easier. [4]

**Sources**
1. [THP Strength – World Record Vertical Jumper's Exact Warm-Up Routine (50.
5 Inches) @ 00:51](https://www.youtube.com/watch?v=O4piVLyfvSA&t=51s)
2. [THP Strength – Do These Stretches To Jump Higher @ 01:24](https://www.yo
utube.com/watch?v=GKjIt1dsgNc&t=84s)
3. [John Evans – If You Have Less Than a 30" Vertical, Watch This @ 12:37](h
ttps://www.youtube.com/watch?v=qPvipeWYWlE&t=757s)
4. [John Evans – A Practical 2 Day Per Week Jump System Guaranteed To Get Yo
u Dunking @ 05:42](https://www.youtube.com/watch?v=1sQ-FOxDP5M&t=342s)

searches: ['static stretching before jumping warm-up', 'long held stretches
reduce power output jump performance']
citation check passed: True | Verifier: pass
```

2. Safety check (code, no LLM). Red-flag symptoms get a physio referral before any agent runs; plan requests are routed to the Planner.

In [3]:
```
from src.red_flags import referral_kind
from src.router import route
```

In [4]:
```
for m in ["Should I static stretch before jumping?", "I felt a pop in my kne
          "Make me a plan for the next 12 weeks"]:
    print(f"{m:42} referral={referral_kind(m)}  route={route(m)}")
```

```
Should I static stretch before jumping?    referral=None  route=coach
I felt a pop in my knee this week          referral=physio  route=coach
Make me a plan for the next 12 weeks       referral=None  route=program
```

3. First plan from the Planner. Needs a profile first (`uv run python -m scripts.plan profile`). The Planner drafts a plan, the code checks and Verifier run, then the graph pauses for my approval.

The Planner's LangGraph setup (`build_plan_graph` in `src/planner/graph.py`). `context` loads the profile and current plan, `planner` ⇄ `tools` is the LLM tool loop, `review` runs the 8 code checks and the Verifier (failures

go back to the planner, at most 2 retries), `approve` pauses the graph until I approve, reject or revise, and `save` writes a new plan version.

The athlete profile (me). This is what the Planner reads with its `get_profile` tool. It's stored as `profile.json` and made with `uv run python -m scripts.plan profile`.

In [5]:
```
from src.planner.store import load_profile

print(load_profile().model_dump_json(indent=2))
```

```
{
  "training_age_years": 4.0,
  "lifts": {
    "back_squat_lb": 225.0,
    "half_squat_lb": 245.0,
    "power_clean_lb": 175.0
  },
  "standing_vertical_in": null,
  "approach_vertical_in": 32.0,
  "standing_reach_in": 86.5,
  "height_in": 67.0,
  "days_per_week": 7,
  "available_days": [
    "mon",
    "tue",
    "wed",
    "thu",
    "fri",
    "sat",
    "sun"
  ],
  "session_minutes": 90,
  "equipment": [
    "barbell",
    "rack",
    "platform",
    "box",
    "med ball",
    "safety bar",
    "belt squat",
    "bench",
    "hamstring curl machine"
  ],
  "event": null,
  "goal": "Dunk off two feet. Approach vertical about 31–33 in, standing rea
ch about 7 ft 2–3 in (needs roughly 10 ft 6 in touch). Back squat is a deep
squat; half squat estimated.",
  "knee_pain": {
    "status": "ache",
    "notes": "Knee pain in the past, now manageable."
  },
  "injury_notes": "Two-foot jumper. No sport or other jump load during the w
eek."
}
```


In [6]:
```
from src.planner import run
from src.planner.render import render_proposal

plan = run.start("new")
print(render_proposal(plan))
```

```
What changes:
  New plan outline
  Month 1: volume
  Month 2: intensification (power and 6–8 rep strength)
  Month 3: absolute strength
  (+5 more)
Why: Proposes a 24-week long-conjugate macrocycle from Mon 2026-11-09 to 202
7-04-25 with six monthly mesocycles. Month 1 week 4 is the unload. Day-by-da
y sessions are written for the first two weeks (Nov 9–22): lifts Mon/Wed/Fr
i, general days Tue/Thu, and a jump session Saturday.
⚠ mesocycles[2]: partly supported — Absolute strength, heavy clean pulls, No
rdics and 2-up-1-down are stated; evidence says about 3 weeks, not 3–4.
```

In [7]:
```
plan = run.resume(plan.thread_id, "approve")  # or "reject", or "revise" wit
print(render_proposal(plan))
```

```
Saved as version 1.
```

The saved plan in detail. The 6-month outline and every week written so far (the Planner writes 2 weeks at a time and rolls forward), then the videos the plan draws on. Every session cites the transcript chunks it came from; those are hidden here and shown in the Developer view.

In [8]:
```
from src.kb.evidence import chunk_meta
from src.planner.render import render_outline, render_week
from src.planner.store import load_program
from src.planner.templates import TEMPLATE_METHOD

program = load_program()
print(f"Plan version {program.version}: {program.change_summary}\n")
print(render_outline(program) + "\n")
for week in program.weeks:
    print(render_week(program, week.start) + "\n")

cites = {c for m in program.mesocycles for c in m.cites} | {
    c for w in program.weeks for d in w.days for s in d.sessions for c in s.
videos = {}  # one line per video the plan draws on
for meta in chunk_meta(cites, method=TEMPLATE_METHOD).values():
    videos.setdefault(meta["title"], meta["url_t"])
print("Videos used:")
for title, url in sorted(videos.items()):
    print(f"  {title}  {url}")
```

Plan version 1: Proposes a 24-week long-conjugate macrocycle from Mon 2026-11-09 to 2027-04-25 with six monthly mesocycles. Month 1 week 4 is the unload. Day-by-day sessions are written for the first two weeks (Nov 9–22): lifts Mon/Wed/Fri, general days Tue/Thu, and a jump session Saturday.

Month 1 · Mon Nov 9 – Sun Dec 6 · volume · Build work capacity with general work that looks little like jumping; week 4 (Nov 30–Dec 6) is the unload.
Month 2 · Mon Dec 7 – Sun Jan 3 · intensification (power and 6–8 rep strength) · Intensify the month-1 volume with clean pulls and power cleans; accessories prime month 3.
Month 3 · Mon Jan 4 – Sun Jan 31 · absolute strength · Heavy squats and heavy clean pulls; eccentric accessories prime month 4.
Month 4 · Mon Feb 1 – Sun Feb 28 · strength-speed (fast bars, eccentric overload) · Power clean to maxes; accentuated eccentric squats and split squats; general days get more general.
Month 5 · Mon Mar 1 – Sun Mar 28 · speed-strength (bar speed) · Move bars fast (lighter cleans, power snatch, hang cleans); a heavy squat primer plus high-velocity squats; start low-volume shock loading.
Month 6 · Mon Mar 29 – Sun Apr 25 · specificity · Highest specificity, the most plyo intensity and variety of the year; accessories become general and slow.

Month 1 of 6 · volume · Week 1 of 4

Mon power clean 5×4 · back squat 4×10 @ 65% · RDL 3×10
      power clean: Power exercise first; crisp, fast reps at a load that moves well.
      back squat: About 145 lb (65% of 225 lb); deep squat, controlled.
      RDL: Hip hinge; general hamstring work that does not mimic jumping.
Tue general circuit (varied movements)
      general circuit (varied movements): Bodyweight or bodybuilding circuit with many different movements; easy and optional.
Wed power clean 5×4 · back squat 4×8 @ 68% · hip thrust 3×10
      power clean: Power exercise first.
      back squat: About 153 lb (68% of 225 lb).
      hip thrust: Heavy loading with little risk; hips rather than knees.
Thu general circuit (varied movements)
      general circuit (varied movements): Different movements from Tuesday; keep it easy.
Fri power clean 5×4 · back squat 4×6 @ 70% · lying hamstring curl 3×12
      power clean: Power exercise first.
      back squat: About 158 lb (70% of 225 lb).
      lying hamstring curl: Barely taxing; finish the session with it.
Sat jump session warm-up jumps @ 35 min · jump session max-effort jumps @ 5 min
      jump session warm-up jumps: Low-effort jumping inside the 40-minute window.
      jump session max-effort jumps: Max-effort jumping at the end.

ⓘ sources

Month 1 of 6 · volume · Week 2 of 4

Mon power clean 5×4 · back squat 5×10 @ 65% · lying hamstring curl 3×12
      power clean: Power exercise first; add load only if the bar stays fast.

back squat: About 145 lb; one more set than last week.
lying hamstring curl: Easy accessory at the end of the session.
Tue general circuit (varied movements)
general circuit (varied movements): Many different movements; easy and optional.
Wed power clean 5×4 · back squat 5×8 @ 68% · Nordic hamstring curl 3×8
power clean: Power exercise first.
back squat: About 153 lb.
Nordic hamstring curl: Eccentric emphasis; stop short of failure.
Thu general circuit (varied movements)
general circuit (varied movements): Different movements again; keep it easy.
Fri power clean 5×4 · back squat 4×6 @ 70% · hip thrust 3×10
power clean: Power exercise first.
back squat: About 158 lb.
hip thrust: Hips rather than knees.
Sat jump session warm-up jumps @ 35 min · jump session max-effort jumps @ 5 min
jump session warm-up jumps: Low-effort jumping inside the 40-minute window.
jump session max-effort jumps: Max-effort jumping at the end.

ⓘ sources

Videos used:
3 Best Exercises To Jump Higher In 2025 https://www.youtube.com/watch?v=OAsUemGRrpw&t=891s
Everything Athletes Ask Us About Jumping Higher https://www.youtube.com/watch?v=3vmd30VvzRA&t=855s
How To Train Your Hamstrings To Jump Higher https://www.youtube.com/watch?v=I1baSIlV_pQ&t=408s
How to Get a 40" Vertical (Full Blueprint) https://www.youtube.com/watch?v=Icml891MLHA&t=209s
My Full Weekly Vertical Jump Training Routine Explained https://www.youtube.com/watch?v=SXjO279Oslw&t=137s